Ivan Evdokimov, Ph.D.

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Professional Summary

Software Engineer with 4+ years of production experience building backend systems, cloud infrastructure, and ML pipelines. Proven track record delivering across the full stack — from PostgreSQL-backed APIs and CI/CD automation to AWS-based MLOps infrastructure — with 95%+ model accuracy on sensitive datasets. Comfortable working across engineering, research, and client-facing teams to deliver reliable, production-grade systems.

Languages: English, French, Chinese, Russian.

Experience

Software Engineer – Full Stack & MLOps, University of Essex — Colchester, UK — Nov 2025 — Present

- Designed and implemented GitHub Actions and Vercel CI/CD pipelines for an AI-assisted development workflow, enabling automated testing, validation, and weekly release cycles across a multi-team internal platform.
- Built end-to-end MLOps infrastructure for an internal platform serving multiple engineering and research teams, reducing model deployment time and enabling continuous monitoring across production environments.
- Full-stack developed a native mobile application for an engineering client, enabling 20 field technicians to submit inspection reports via photos, recordings, and text; integrated the OpenAI API to automatically generate well-formed reports written directly to a database, saving 3 weeks of manual data collection annually and improving client communication.

Software Engineer – Backend & ML, UK Data Service — Colchester, UK — Jan 2023 — Oct 2025

- Trained, fine-tuned, and deployed domain-specific PyTorch and scikit-learn models using Hugging Face, vLLM, and Ollama for automated metadata classification and disclosure risk assessment on a public data portal, achieving 95%+ accuracy on sensitive social science datasets.
- Applied SHAP (SHapley Additive exPlanations) across production models to identify and select the most predictive features and to provide interpretable, audit-ready explanations of classification decisions on sensitive data.
- Built ETL pipelines to extract, transform, and query PostgreSQL-hosted social science datasets of 100K+ records as model inputs for automated metadata classification, improving throughput and classification accuracy at scale.
- Optimized statistical disclosure control algorithms by migrating from R to C++ using bitmask techniques, achieving 11x performance improvement and enabling real-time processing of datasets with 500k+ records.
- Collaborated with academics, project managers, and public sector clients through regular requirements meetings, translating diverse non-technical needs into production ML system specifications.

Research Officer & ML Instructor, University of Essex — Colchester, UK — Oct 2021 — Dec 2022

- Migrated quantitative macroeconomic models from MATLAB to C++, reducing simulation runtime 8x for collaborations with other universities in the UK and the USA.
- Delivered lectures and labs in Data Structures, C/C++, and Machine Learning to 100+ undergrad and postgrad students, achieving an average student feedback score of 4.8/5.

Projects

Chat-LLM — github.com/ivan020/ChatLLM

- Built a self-hosted RAG pipeline over a Raspberry Pi-hosted Ollama LLM, indexing past stream transcripts so viewers could query historical content and prior statements in real time via Twitch chat.
- Optimised the Qwen3-Coder-480B for low-latency inference under constrained hardware.

FinBot: Chat-Based Trading Simulation — github.com/ivan020/FinBot

- Built a real-time portfolio tracker in Go with buy/sell mechanics, a PostgreSQL (Supabase) backend recording all transactions, and a live leaderboard — using Twitch chat as the user interface.
- Developed a TypeScript React frontend with real-time portfolio visualisation, interactive performance graphs, and transaction history, serving 120+ monthly users.

MTG Card Reader — ivan020.github.io/MtgRecognizer/

- Built a Golang REST API that uses OCR to extract text from Magic: The Gathering card photos, identifies the card via LLM, and returns real-time price data from a PostgreSQL-backed catalogue.
- Developed a TypeScript React frontend with card filtering, deck management, and real-time price display, deployed on Vercel.

Education

University of Essex, PhD in Computational Finance — Oct 2021 — May 2025

- Developed a PyTorch transfer learning framework — incorporating RNN and MLP architectures — for Bayesian model averaging in financial forecasting; applied SHAP to interpret model outputs and identify the most influential predictive features.
- First-authored a peer-reviewed CS/ML journal publication applying SHAP-based explainability analysis, and co-authored 3 international conference papers; released an open-source package with external adoption.

University of Essex, MSc in Financial Econometrics — Oct 2020 — Sep 2021

- Sole-authored a modular C++ agent-based simulation modelling household consumption, employment, and borrowing behaviour under dynamic and negative interest rates, with an extensible architecture supporting additional firm and household modules.
- Presented MSc thesis findings and simulation results at an academic conference, demonstrating research communication skills to an international audience.